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FENG 498 PROJECT REPORT



WAREHOUSE LAYOUT OPTIMIZATION

A Simulation Study of Storage Assignment Policies
at the Schneider Electric Manisa Plant Warehouse

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Abstract

This study presents a discrete-event simulation model built to evaluate storage-assignment policies at the Schneider Electric Manisa medium-voltage switchgear plant, with the goal of improving the efficiency of its kitting operation. The work was motivated by a limitation in the plant’s existing ABC/FMR classification tool, which ranks materials by aggregate demand and ignores which production line a material is kitted for. As a result, items that belong to the same kit are scattered across different corridors, which creates queues at the shared reach-truck (RT) fleet and delays material supply. A SimPy-based model was developed from four months of historical SAP dispatch data and calibrated with a video time-motion study, and six storage-assignment policies were compared across five key performance indicators: kitting lead time, operator waiting time, reach-truck utilization, throughput, and operator walking distance. Across $N = 20$ paired replications, the proposed *Line-aware Slotting* policy reduced mean kitting lead time from 12.37 to 6.91 minutes per order, a reduction of about 44% ($p < 0.001$), and lowered reach-truck utilization from 21.7% to 1.3%. Walking distance rose only slightly, from 89.0 to 92.4 metres per order, which shows that the improvement comes from removing reach-truck calls from the kit path rather than from making operators walk less. One-way ANOVA and common-random-number paired t -tests confirmed that these differences are statistically significant and not an artefact of simulation noise. The study shows that a meaningful gain is available from re-slotting alone, without new equipment or physical layout changes.

1 Introduction

Warehouse optimization is the systematic improvement of facility operations toward efficient, flexible, and cost-effective performance. It draws on several levers, including workflow design, technology, spatial layout, and inventory management, and its aim is to reduce operating cost while improving speed, accuracy, and productivity. Among warehouse activities, order picking is the most labour-intensive and costly process, and roughly half of all picking time is spent travelling between storage locations [9]. Storage location assignment, that is deciding which material is placed in which bin, is therefore one of the few levers that can improve picking efficiency without significant capital expenditure [5].

Schneider Electric is a global company in energy management and automation with operations in more than 100 countries. Its Manisa medium-voltage switchgear plant runs a pallet-rack warehouse that feeds several parallel production lines through a coordinated kitting process. Kitting here means assembling the complete set of components for a specific production order, a kit, before assembly begins. The warehouse therefore dispatches coordinated batches of components, one batch per order, rather than single parts.

The facility handles both pallet-based and box-based materials. It uses multi-level rack systems, dedicated fast-mover shelves, and kitting corridors that feed the production lines. Material

flow is maintained by a milkrun (waterspider) system on fixed 45-minute cycles, using reach trucks (RT) and electric pallet trucks (EPT). These cycles impose tight time limits with little tolerance for delay. The warehouse also operates under strict safety rules, including unidirectional flow and dedicated kit paths, which restrict routing options and increase the impact of slotting decisions.

The scale of the operation makes storage assignment important. The warehouse processes roughly 400 kit orders on an active working day; the SAP ZWM92 dispatch log covering 2 January to 18 May 2026 records 40,804 kit orders across 103 active days. At this volume, a one-minute reduction in mean kitting lead time saves the equivalent of more than six working hours of aggregate delay each day.

Despite these stakes, the current storage-assignment practice, driven by a capacity heuristic and a traditional ABC/FMR classification, does not account for the production line a material is kitted to. High-frequency items are spread across different corridors, which produces queues at the reach-truck fleet and delays material supply. Physically re-slotting thousands of pallet positions to test alternatives would be expensive and disruptive. This study therefore develops a validated discrete-event simulation (DES) model to evaluate candidate dependency-aware slotting policies against four months of real demand, so their operational impact can be measured without moving a single physical pallet.

1.1 Problem Statement

The warehouse at the Schneider Electric Manisa plant operates in a dynamic environment where pallet storage, kitting, and internal transport all take place at the same time and depend on the same shared resources. Within this environment, several operational and capacity-related issues in the current structure limit its scalability, flexibility, and efficiency, and they are the direct subject of this study.

The facility is a multi-level rack system built mainly for standard Euro pallets (80×120 cm), with a total capacity of about 3,000 pallet positions distributed across several modules, and each rack section is structured to hold three pallets. The layout is organized into parallel aisles roughly three metres wide, where handling is carried out mostly by reach trucks. This narrow-aisle arrangement creates a strong dependency on a small number of shared vehicles and raises the risk of congestion and delay during busy periods. For safety, access to most standard racks is restricted to a single side, with the rear closed to prevent contact between operators and vehicles. Frequently used materials are instead stored at ergonomically accessible levels on dedicated fast-mover racks with double-sided access: replenishment is performed from the front by reach truck while operators pick from the rear through the kitting aisle, so that picking and replenishment can proceed at the same time without the two streams interfering.

Materials enter the warehouse through the receiving process and are assigned to storage based on stock level and day-to-day operational decisions. As a rule, high-frequency items are

placed on the fast-mover racks while less frequent items are sent to the upper levels. Retrieval for production is carried out either through coordinated kitting or through direct pallet handling, and internal transport is managed by the milkrun system, which follows predefined routes to feed the assembly lines at fixed intervals.

Despite this structured physical system, the current storage and allocation policies do not accurately reflect how materials are actually used across the different production lines. The existing classification relies on an internal combined ABC/FMR tool, in which the ABC component represents material cost and the FMR component represents usage frequency. Although the tool is refreshed quarterly and flags materials that are no longer or only rarely used, it carries a fundamental flaw: it treats every material as if it originated from a single homogeneous source and ignores the differing quantities that each manufacturing line consumes. As a result, the raw output is not reliable for direct operational use. The plant has introduced a workaround that analyses per-line kit consumption, but the overarching system still assigns storage locations by aggregate usage frequency, without accounting for the operational dependencies between a material and the line it serves. This simplification leads to suboptimal placement decisions and frequently scatters the materials belonging to a single kit order across entirely different rack levels and zones [6, 17].

This scattering is the root of the operational problem. When the components of one kit are spread out, with some sitting in accessible fast-mover positions and others placed in high-level locations that require a reach truck, the manual operator in the kit corridor cannot complete the kit alone. The operator must stop and wait for a reach-truck driver to retrieve the high-level items, which produces a characteristic "waiting waste" during order preparation. Because the seven reach trucks are a shared resource, these requests queue, and the waiting time propagates into kitting lead time and reduces synchronization between the warehouse and the production lines. The impact is amplified by the timing of the operation: the milkrun runs on 45-minute cycles with only a narrow 20% stock tolerance at each station, so a delay in kit preparation can quickly exhaust the buffer and threaten a line stoppage [17]. At a volume of roughly 400 kit orders per active day, even a small per-order delay accumulates into hours of aggregate waiting across the shift.

The layout also does not fully exploit the available space. Storage density could in principle be raised through layout redesign, additional racking, or better use of vertical space, but such physical changes are costly and disruptive and require quantitative assessment before they are attempted. At present the plant follows predefined internal rules that have never been systematically compared against alternative strategies under consistent, controlled conditions, so there is no objective evidence about how much could be gained from a better assignment policy.

This study therefore targets the inefficiency that arises from the absence of a structured, dependency-aware approach to storage assignment. Rather than proposing immediate physical modifications, it evaluates improvement alternatives within a virtual environment, which allows a low-cost assessment of how policy changes would affect operations before any physical im-

plementation. The alternatives range from established methods in the literature, such as Double ABC and time-based classification, to a strategy tailored to the specific operational characteristics of the plant, and all of them are measured against the current baseline under identical demand and resource conditions.

1.2 Motivation

Warehouse efficiency is the main driver of production stability at this plant. The entire material flow is governed by a strictly timed milkrun system that runs on 45-minute cycles and completes nine tours per day to replenish the production lines. Each station must hold enough stock to last one 45-minute interval plus a narrow 20% tolerance margin. Within these rigid limits, even a small inefficiency in retrieval or kit preparation can exceed the buffer and lead to an immediate disruption or a costly line stoppage. Beyond the purely operational metrics, occupational health and safety is a fundamental design constraint, and the project is strongly motivated by the need to raise system throughput while strictly maintaining the separation between pedestrian operators and vehicles such as reach trucks and forklifts.

The facility is operationally mature and is supported by an established culture of optimization. A previous internal study reduced the fleet by one reach truck while maintaining the same performance level, which is direct evidence that meaningful gains can be achieved through better system design rather than through capital expenditure. The warehouse currently manages its operations with a diverse fleet of 7 reach trucks, 3 forklifts, 7 milkrun trains, and 2 electric pallet trucks, together with the Kardex units used for the efficient handling of small parts.

Despite this advanced infrastructure, critical inefficiencies persist. The primary analytical weakness is the existing classification tool, which calculates its results as if all materials belonged to a single homogeneous group and therefore ignores the product complexity and the destination line of each material. Operationally, this misalignment creates significant waiting waste during order preparation. The materials for a single kit order are frequently scattered: some sit in accessible fast-mover positions while others are placed in high-level locations that require a reach truck. This produces a severe bottleneck in which a manual operator in the kit corridor must stop and wait for a reach-truck driver to retrieve the high items, which extends lead time and reduces the synchronization of the whole process.

The motivation of this study is therefore to provide a scalable and cost-efficient solution to these challenges through a data-driven slotting approach. By combining a refined, dependency-aware value-based (ABC) and frequency-based (FMR) analysis, the work aims to position high-frequency materials in ergonomic, lower-level locations so that they can be picked by hand quickly, which removes the reach-truck dependency from the kit path and attacks the waiting waste at its source.

1.3 Research Objectives and Scope

The objectives of this study are:

1. to develop and validate a SimPy-based discrete-event simulation model of the Schneider Electric Manisa warehouse, driven by four months of SAP dispatch data and calibrated with a video time-motion study;
2. to design and implement six storage-assignment policies, including the proposed *Line-aware Slotting* policy, within a single simulation framework that holds the demand stream and resource pool constant;
3. to compare the policies across five key performance indicators (kitting lead time, operator waiting time, reach-truck utilization, throughput, and operator walking distance); and
4. to recommend a placement strategy with statistical evidence, using one-way ANOVA, common-random-number paired t -tests, and 95% confidence intervals from $N = 20$ replications.

The scope is limited to storage location assignment for the kitting (order picking) process. The physical layout, the resource fleet, and the milkrun schedule are held fixed. Replenishment scheduling, workforce planning, dynamic routing, and equipment investment are outside the scope, so that the effect of slotting alone can be isolated.

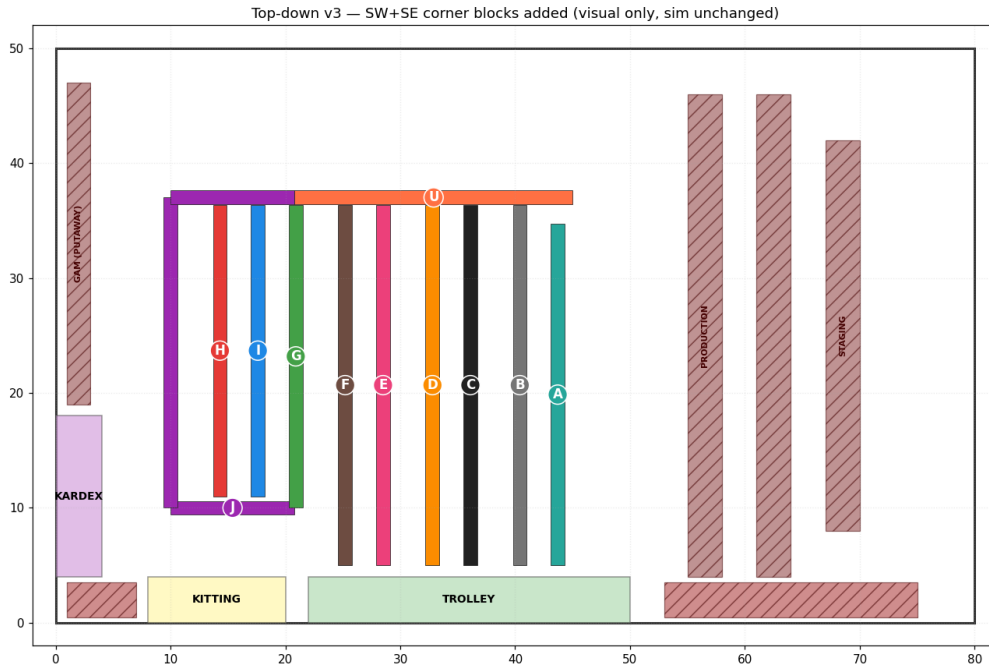


Figure 1: Top-down layout of the modelled warehouse. Rack rows are labelled A through J and U; kit corridors (about 2 m) alternate with reach-truck aisles (about 3 m) along the east-to-west sequence.

2 Literature Review

This section reviews work across four threads: discrete-event simulation (DES) in warehousing, warehouse layout optimization with multi-criteria classification, automated storage and retrieval systems (AS/RS), and decision support systems (DSS) in warehouse management. It closes by identifying the research gap that motivates the proposed line-aware slotting policy.

2.1 Simulation-Based Warehouse Analysis

Simulation-based approaches are frequently used to evaluate warehouse performance and capacity-related issues in a comprehensive way. Discrete-event simulation (DES) makes it possible to analyse complex systems that involve stochastic arrivals, variable processing times, and constraints on shared resources. Negahban and Smith [18] demonstrate that simulation is an effective tool for identifying bottlenecks, evaluating storage utilization, and testing alternative system configurations in both manufacturing and warehouse environments. Such approaches are particularly useful for questions like whether the shipment area is large enough or how the workforce should be allocated, and they can be explored without disrupting the real operation. Hapsari et al. [12] reach a similar conclusion for an Indonesian pharmaceutical warehouse, where a discrete-event model surfaced bottlenecks and inventory-control problems that a static ABC analysis could not reveal. A recurring limitation, however, is that many studies focus on a single aspect such as routing or capacity in isolation, whereas a real warehouse requires an integrated analysis that brings together layout, routing, and operational decision-making.

2.2 Warehouse Layout Optimization

Ardiansyah et al. [2] address the operational inefficiencies in warehouse picking that are caused by unoptimized layouts and irregular product placement, and they note that picking typically consumes far more time than checking, packing, or loading. Their case study identifies how large-volume items, such as expired motorcycle tires, can obstruct forklift lanes and exceed the racking capacity, which leads to a breakdown of First-In-First-Out (FIFO) discipline. To resolve these problems, the authors integrate Double ABC analysis, which classifies inventory by both pickup frequency and outer volume, with discrete-event simulation built in Arena to model and validate the outbound process. A one-way ANOVA confirmed a significant reduction in moving time, showing that the proposed layout reduces operator travel distance and helps locate the bottlenecks in the system. Their work provides a mathematically validated framework for optimizing limited-space warehouses, and it is the methodological basis for the Double ABC policy evaluated in the present study.

2.3 Automated Storage and Retrieval Systems (AS/RS)

Automation technologies such as AS/RS are often proposed to improve warehouse performance. AS/RS use cranes or robotic systems in high-density racks, which increases space efficiency and reduces dependence on manual labour. The literature reports advantages including higher storage density from narrower aisles, shorter travel time, better accuracy, and safer operation through reduced human-vehicle interaction. These systems also carry trade-offs, mainly high investment cost and reduced flexibility [20].

AS/RS design involves two parts: physical design and control policies [20]. Physical design fixes rack dimensions, aisle configuration, and density; control policies set the storage-assignment strategy, such as dedicated, random, or class-based storage. Class-based storage is widely preferred because it balances accessibility against space use. Performance is usually judged by travel time, throughput, waiting time, and resource utilization. AS/RS represents the highest level of physical automation, but the present study shows that meaningful gains are available from slotting policy alone, without the capital cost of full automation.

2.4 Decision Support Systems in Warehouse Management

Warehouse design, storage location assignment, and order picking are frequently studied as separate problems, even though in practice they are closely linked. Moghadam et al. [7] show that treating these elements independently leads to suboptimal performance, because layout choices directly determine where items are stored and how efficiently they can be picked. The present study addresses this by evaluating different storage and classification policies without physically altering the warehouse layout, an approach supported by Amorim-Lopes et al. [1], who show that combining simulation with optimization can yield significant reductions in travel distance and order-retrieval time even when the physical layout remains unchanged.

To support such integrated evaluations, decision support systems (DSS) are increasingly used in warehouse management to drive data-driven decisions and improve operational efficiency. In one representative case study, a warehouse-management DSS was developed to analyse operational data and help managers evaluate alternative scenarios for layout design, material flow, and resource utilization; by integrating several performance indicators, the system enabled a systematic assessment of operations and replaced purely intuitive managerial judgement. The broader literature indicates that DSS-based approaches improve the visibility of warehouse processes, raise decision quality, and help reveal hidden inefficiencies in material handling and storage. These studies, however, tend to emphasize high-level decision-making frameworks rather than a detailed model of the underlying operations. Combining the DSS philosophy with analytical methods such as simulation therefore gives a more complete solution. That is the approach taken here: the operations are modelled in detail first, which produces a statistically validated, data-driven basis that could later be integrated into a broader decision-support framework.

2.5 Research Gap and Contribution

The literature covers storage assignment, simulation-based analysis, and layout improvement extensively, but most studies evaluate materials by aggregate demand and treat items as independent [10, 17]. In synchronized, production-driven warehouses, however, material flow is shaped by the requirements of specific production lines and by kitting: components are consumed in coordinated sets, so their relative locations directly affect kit preparation time [11, 19].

These system-specific dependencies are rarely addressed in the reviewed literature, where classification methods are generally applied in a uniform way without being adapted to the operational characteristics of a particular warehouse [13, 20]. The critical gap, then, does not lie in the absence of established assignment methods, but in the failure to adapt and evaluate those methods within a real-world warehouse context that explicitly accounts for production-line-based consumption variability and kit-level spatial dependencies.

Addressing this gap, the primary contribution of this study is the development and evaluation of a dependency-aware classification approach, the proposed *Line-aware Slotting* policy, assessed against five alternative policies within a single discrete-event simulation framework. The study examines how established storage strategies can be adapted in practice to reflect actual operational realities rather than theoretical aggregate demand. By evaluating all of the policies against the current baseline under identical demand and resource conditions, the research provides a statistically validated, application-oriented methodology [2, 15]. The result is a scalable approach for reducing waiting waste and improving material synchronization in complex, production-driven environments, achieved without any physical layout modification.

3 Methodology

This section describes the framework used to analyse the current operation, develop alternative storage and classification policies, and evaluate their impact. The approach is built on historical SAP data, detailed system modelling, and discrete-event simulation. By replicating the warehouse in software, the policies are tested under identical, controlled conditions, which allows a fair comparison against fixed metrics. Figure 2 summarizes the five phases of the work.

3.1 Data Collection

The data came directly from the company’s SAP system, in cooperation with company representatives. The analysis uses operational data from 2 January to 18 May 2026, covering 40,804 kit orders across 103 active working days.

The primary dataset is the SAP ZWM92 transaction log, which records material withdrawals, transaction quantities, and the flows associated with kitting and production support. These records were combined with withdrawal quantities to determine the actual usage frequency of each component. During preprocessing, records with zero or negative withdrawal

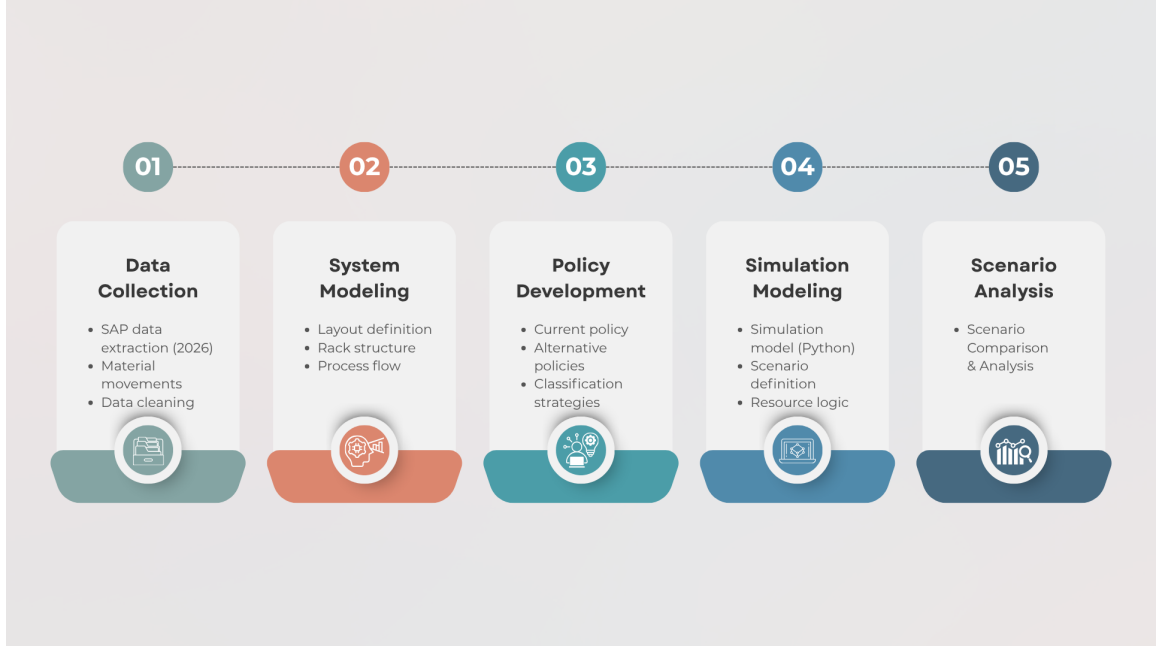


Figure 2: The five-phase methodology framework for the simulation-based warehouse study.

quantities were removed to keep the dataset clean.

Several supplementary datasets were added: current storage bins, per-line material usage, and the company's existing ABC/FMR results. An independent Pareto analysis was carried out on total consumption: materials up to 80% of cumulative consumption were classified as A, the next 15% (to 95%) as B, and the remaining 5% as C [8, 16], as defined in Equation (1). This classification was then compared against the existing one to document differences.

$$\text{Class}(i) = \begin{cases} A, & \text{if } P_i \leq 0.80, \\ B, & \text{if } 0.80 < P_i \leq 0.95, \\ C, & \text{if } P_i > 0.95, \end{cases} \quad P_i = \frac{\sum_{j=1}^i c_j}{\sum_{j=1}^n c_j}, \quad (1)$$

where c_j is the consumption of material j ranked in decreasing order and P_i is the cumulative consumption share up to rank i .

Technical layout drawings and rack documents supplied the physical dimensions: rack modules, storage levels, fast-mover shelves, aisle widths, and the relationships between storage and kitting zones. Bill of Materials (BOM) data and historical kitting records were used to map material relationships and kit-based usage [4].

Beyond the SAP data, operational parameters were collected through field observation. A video time-motion study recorded 2,319 micro-events on the F400 production line, capturing operator walking speeds, reach-truck movement and lifting times, manual picking durations, and the operator waiting time caused by reach-truck dependency. Operator allocation, fast-mover usage, reachable levels, and handling patterns were recorded as well. Together these sources formed the quantitative basis for policy development, modelling, and evaluation.

3.2 System Modelling

The warehouse is represented within a discrete-event framework. Storage zones are modelled as distinct locations; operators and reach trucks (RTs) are modelled as active resources; the milkrun trains are transport entities; and production orders are kit-preparation requests triggered by demand.

The physical layout has 11 rack modules with a total capacity of 3,203 pallet positions. Materials are unitized on standard Euro pallets (80×120 cm). Reach-truck aisles measure 300 to 320 cm and dedicated kit corridors are 200 cm wide, as confirmed by the facility AutoCAD measurements. Rack heights reach about 7.5 m, which requires vertical storage handling [4].

A central design feature is the separation between standard storage and fast-mover sections. The fast-mover zone, the three lowest rack levels up to about 206 cm, is the manual band, where operators can reach materials without a reach truck. Here the racks allow decoupled work: operators pick from one side while replenishment is performed from the other, which removes direct contact between personnel and equipment. Standard and upper-level positions require reach-truck support for all retrievals [20].

The current state follows a hybrid logic in which assignments are influenced by line requirements, capacity heuristics, and the existing ABC/FMR classification. In practice, operations deviate from these rules because of space limits and immediate replenishment needs, which fragments placement across zones. This fragmentation adds travel distance, increases reach-truck dependency, and creates waiting time during kit preparation, especially for materials in upper levels. The model captures these interactions when assessing the alternative policies [10].

3.3 Policy Design

This section describes the storage and classification policies evaluated in the study. The policies use different elements of the operational data: physical accessibility, usage frequency, retrieval time, and production-line and kit dependencies.

3.3.1 Baseline and Capacity Heuristic (Control Group)

Two reference policies form the control group:

- **Baseline (Actual SAP).** This policy reproduces the historical placement of the warehouse, reflecting patterns accumulated over years of operation and daily replenishment. It is the main reference point for measuring the real-world impact of the alternatives [15].
- **Capacity heuristic.** Materials are assigned to the first available slot that meets the item's storage requirement, with no line-aware logic or demand dependency. Comparing the proposed policy against this heuristic isolates the gain from dependency-aware slotting versus simple space filling [20].

3.3.2 Double ABC Analysis Policy

The Double ABC policy follows the method of Ardiansyah et al. [2], which shows that combining two classification dimensions gives a more granular prioritization than a single-axis ABC. Because reliable material-volume data were not available, the second dimension uses material cost rather than volume. The first dimension captures economic importance (cost) and the second captures operational importance (frequency). The two criteria are combined in Table 1.

Table 1: Double ABC classification matrix.

Dimension	Classes	Thresholds
Cost (C)	High / Medium / Low	by unit cost
Frequency (F)	High / Medium / Low	> 80% / 80–95% / 95–100% cumulative

The intersection of the two criteria gives a 3×3 matrix with nine categories (AA through CC):

- **AA (high cost, high frequency):** highest priority, assigned to prime locations such as fast-mover shelves and lower levels.
- **CC (low cost, low frequency):** lowest priority, assigned to upper levels or less accessible locations.
- **Intermediate (AB, BA, and so on):** placed by their combined cost and frequency, balancing accessibility against inventory value.

This moves the plant beyond simple frequency-based classification and prevents rack capacity from being consumed by bulky, low-demand items.

3.3.3 Time-Based Classification Policy

Unlike static methods based on frequency or volume, the time-based policy uses the accessibility of each location. In a high-bay warehouse, total retrieval time is dominated not by horizontal travel but by vertical lift and the need for reach-truck intervention [4, 14]. The policy estimates an access time for each location from travel speed, lift speed, and reach-truck handling, then assigns frequently used materials to the locations with the shortest access time. Table 2 summarizes the mapping. By using time rather than distance alone, the policy aims to reduce lead time and waiting and to support line synchronization [19, 17].

Table 2: Time-based policy: storage zone assignment by access time.

Relative access time	Storage assignment	Access method
Minimal	Fast-mover shelves	Manual picking
Short	Lower rack levels	Reach truck, minimal vertical travel
Moderate	Mid rack levels	Reach truck, standard vertical travel
Extended	Upper levels	Reach truck, significant vertical travel

3.3.4 Line-aware Slotting (Proposed Policy)

The proposed *Line-aware Slotting* policy is the core contribution of this study. It shifts the focus from warehouse-wide averages to production-line dependencies. The implementation has two stages. First, an independent ABC analysis is run for each production line rather than for the warehouse as a whole, so materials are ranked by their consumption within each line, with A-class defined as the top 80% of cumulative consumption. This removes the distortion caused by aggregated data and reflects the true criticality of each material for its line. Second, the policy uses historical kitting and BOM data to identify items that are frequently picked together and places them in the same rack module or aisle. Materials shared across several lines are positioned centrally in high-accessibility zones to reduce unnecessary travel [11, 17].

The placement logic prioritizes the manual band, the three lowest levels that operators can reach by hand. By keeping high-frequency, line-specific materials in this zone, the policy achieves two goals at once: proximity, so materials sit close to the line they serve, and reach-truck decoupling, so frequent picks no longer require an RT [5]. This directly attacks the waiting waste in the kit corridors and reduced reach-truck utilization to 1.3% in the simulation, as reported in Section 4.

3.4 Simulation Model

The model was built with the Python SimPy library to represent the interaction of warehouse operations and to evaluate the alternative policies. Simulation was chosen because it can capture resource congestion and stochastic arrivals under realistic conditions without disrupting the real operation [10, 18, 3]. The implementation is organized into 17 source modules that handle the full lifecycle: data loading and SAP ZWM92 bin decoding, warehouse geometry, rectilinear routing with rack-body obstacles, slotting policy management, the SimPy process logic, KPI recording, and validation through sensitivity sweeps. Figure 3 shows the lifecycle of one kit order in the model.

Order arrivals are generated from the demand pattern in the SAP ZWM92 log, using empirical inter-batch gaps and batch sizes. The inter-batch arrival gap has a mean of 5.36 minutes and a coefficient of variation of 2.04, while the mean batch size is 4.68 kits. The coefficient of variation of the gap is well above 1.0, which rules out an exponential model and justifies sampling from the empirical distribution rather than a fitted theoretical one [16, 22]. Service times use lognormal distributions parameterized from the F400 time-motion study, with a mean operator pick time of 0.113 minutes, a reach-truck pick-and-place time of 0.110 minutes, a manual high-level pick penalty of 0.102 minutes, and a Kardex retrieval time of 0.113 minutes. The coefficient of variation is defined in Equation (2), and each lognormal distribution is moment-matched to its measured mean and coefficient of variation through Equations (3) and (4).

How the simulation handles one kit order

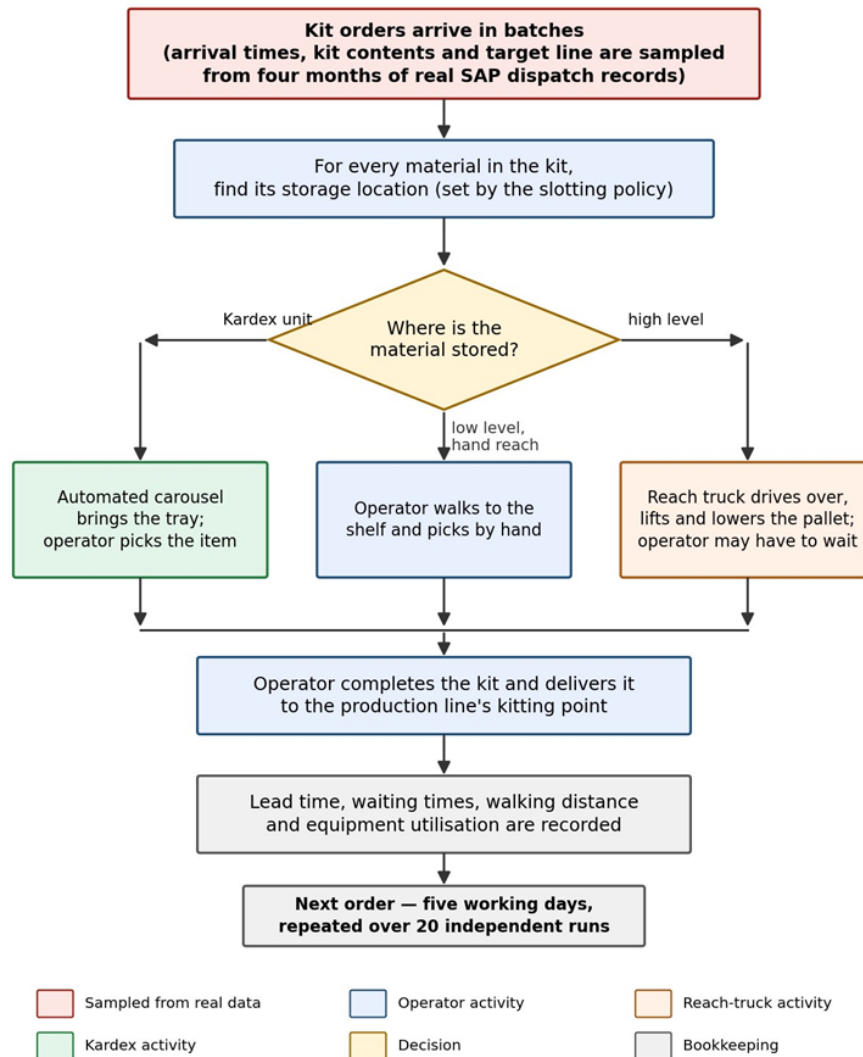


Figure 3: Lifecycle of a single kit order in the simulation.

$$CV = \frac{s}{\bar{x}} \quad (2)$$

$$\sigma = \sqrt{\ln(1 + CV^2)} \quad (3)$$

$$\mu_{\log} = \ln(\bar{x}) - \frac{1}{2}\sigma^2 \quad (4)$$

where $\bar{x}$ and s are the sample mean and standard deviation of the measured time, and σ and $\mu_{\log}$ are the shape and location parameters of the lognormal distribution; the location term in Equation (4) keeps the distribution mean equal to $\bar{x}$ for any σ .

When an order is triggered, the model looks up the storage location of each requested material, computes the shortest obstacle-free rectilinear travel path for the assigned resource, and performs the pick. Travel distance uses the rectilinear (Manhattan) metric of Equation (5), evaluated between pick points and summed over the route; the router searches a set of aisle waypoints and returns the shortest path that avoids rack bodies. If a required resource is busy, the order waits in a queue and the waiting time is recorded.

$$d(A, B) = |x_A - x_B| + |y_A - y_B| \quad (5)$$

The architecture lets the storage-assignment logic be replaced without changing the rest of the model, so every policy faces the same demand stream and resource pool, which keeps the comparison fair. To check spatial accuracy, a 3D viewer was built in Three.js that renders the layout from the AutoCAD DWG file and the eleven rack PDF drawings. Visual inspection of the exported movement paths confirmed that agents follow the corridors and do not pass through rack bodies [2, 21].

3.5 Performance Metrics

The policies are evaluated on five key performance indicators (KPIs):

- **Kitting lead time (min/order).** Time from order creation until the completed kit is ready at the staging area for milkrun pickup. Lower lead time means faster supply to the lines.
- **Operator waiting time (min/order).** Time the kitting operator spends idle, waiting for reach-truck support to retrieve upper-level materials.
- **Reach-truck utilization (%).** Share of total reach-truck capacity actively used for retrieval, defined in Equation (6).
- **Throughput (orders/day).** Number of kits completed per working day, defined in Equation (7).
- **Operator walking distance (m/order).** Total rectilinear path length an operator walks to complete one kit. This metric checks that any lead time reduction comes from smarter

placement, not from making the operator walk further.

$$U_{RT} = \frac{\sum_{r=1}^{N_{RT}} t_r^{\text{busy}}}{T_{\text{sim}} N_{RT}} \quad (6)$$

$$\Lambda = \frac{N_{\text{completed}}}{T_{\text{sim}}} \quad (7)$$

where t_r^{busy} is the busy time of reach truck r , N_{RT} is the number of reach trucks, T_{sim} is the simulated horizon, and $N_{\text{completed}}$ is the number of completed kit orders.

3.6 Verification and Validation

3.6.1 Verification

Verification checked that the model was built correctly and behaves as intended, through code-level checks and logical-flow tests. At the code level, tests covered the SAP bin decoder, the Kardex routing logic, and resource tracking. A ten-step control list confirms that the modelled pallet capacity is consistent with the layout file before any result is recorded, that every material has a valid location, and that pick counts are correct. The model checks that its 3,101 modelled positions reconcile with the 3,203 pallet positions stated in the rack drawings (the difference reflects narrower feeder bays and a cart-parking slot) before a run proceeds. Fixed random seeds make each run exactly reproducible.

The logical flow was then tested step by step, covering order creation, picking, and reach-truck use. The model was also run on small scenarios whose expected travel and waiting times could be computed by hand and compared against the output. Finally, extreme scenarios were tested: with all materials on the lowest levels the model produced zero reach-truck waiting, and with all materials on the top levels it produced a clear rise in waiting and reach-truck use. As Table 3 shows, the difference between the hand calculation and the simulation was below 1% in every scenario.

Table 3: Verification test outcomes.

Test scenario	Expected behaviour	Outcome
Single lower-rack retrieval	Minimal wait, no RT needed	Pass
Single upper-rack retrieval	RT request generated	Pass
No upper-rack materials	Near-zero waiting time	Pass (0.12 min)
All upper-rack materials	High RT use, long wait	Pass

To keep the comparison between policies fair, common random numbers (CRN) were used, so every policy faces the same sequence of orders and any difference comes from the storage strategy rather than from random variation.

3.6.2 Validation

Validation assessed whether the model represents the real warehouse, following the Sargent framework [21] with both operational review and statistical checks. First, the physical and operational elements, including the layout, racks, fast-mover areas, and milkrun cycles, were reproduced from SAP data, drawings, BOM relationships, and field observation, and the process logic was reviewed with company representatives.

Second, a chi-square goodness-of-fit test compared the simulated pick distribution against the historical SAP ZWM92 data. The pooled test gave $\chi^2(7) = 535.05$ ($p < 0.001$), pooled from the ten-rack restricted test $\chi^2(9) = 537.68$ to satisfy Cochran’s rule, with a Cramér’s V of 0.22, which indicates a small-to-medium effect. The statistic and effect size are defined in Equations (8) and (9). Because the simulation assigns pick locations using its own logic rather than replaying history, an exact bin-for-bin match is neither expected nor desired; the rack ranking, however, agrees, and the busiest racks in the log (E, G, and D) are also the busiest in the model. Table 4 summarizes the checks.

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (8)$$

$$V = \sqrt{\frac{\chi^2}{n(k-1)}} \quad (9)$$

where O_i and E_i are the observed and expected pick counts in rack i , k is the number of racks, and n is the total number of picks.

Table 4: Validation summary (Sargent framework).

Validation check	Result	Key metric
Distributional fit (chi-square)	Pass (disclosed)	$\chi^2(7) = 535.05$, $V = 0.22$
Daily volume calibration	Pass	Error 1.1% vs ≈ 396 kits/day
Route model (zero crossings)	Pass	No agent crosses a rack body
Reproducibility	Pass	Stable across seeds

The daily volume calibration matched the real average of about 396 kits per day within 1.1%. The routing model was checked on simulated agent paths, with no agent passing through a rack body. Reproducibility checks confirmed stable results across seeds. Together these checks support the use of the model for comparing storage policies.

3.7 Experimental Design and Statistical Analysis

Each policy was run for $N = 20$ independent replications. Common random numbers were used across policies: replication i of every policy uses seed $42 + i$ for the arrival stream, so all

policies face the same demand trajectory in a given replication. This pairing removes demand-side variance from the policy comparison. Within a policy the replications use distinct seeds and are therefore independent; across policies, the replications are correlated by design through the shared seeds, which is exactly what the paired analysis exploits. A warm-up period was excluded from each run to reduce initialization bias, so only post-warm-up output was retained.

For each replication, the model recorded the mean of each KPI. The sample mean, standard deviation, and a 95% confidence interval were then computed for each policy and KPI. Because $N = 20$, the interval uses the t -distribution with 19 degrees of freedom ($t_{0.025,19} = 2.093$), as in Equation (10).

$$\bar{x} \pm t_{0.025,19} \frac{s}{\sqrt{N}} \quad (10)$$

A one-way ANOVA was used as the primary test, with storage policy as the factor (six levels) and the per-replication KPI as the response, at a significance level of $\alpha = 0.05$. Because each KPI is an average of many events within a replication, its distribution across replications is approximately normal, and the sample sizes are equal, so the assumptions for the parametric test are reasonable. The ANOVA statistic is given in Equation (11) and the hypotheses in Equation (12).

$$F = \frac{MS_B}{MS_W} = \frac{SS_B/(k-1)}{SS_W/(N_{\text{tot}} - k)} \quad (11)$$

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_6 \quad H_1 : \exists i \neq j \text{ such that } \mu_i \neq \mu_j \quad (12)$$

Where the ANOVA was significant, Tukey's Honestly Significant Difference (HSD) test identified which pairs differ. In addition, common-random-number paired t -tests compared each alternative directly against the Actual SAP baseline. The paired test (Equation (13)) uses the per-replication differences, which removes the shared demand variance and gives a tighter contrast.

$$t = \frac{\bar{d}}{s_d/\sqrt{N}}, \quad d_i = \text{KPI}_{\text{alt},i} - \text{KPI}_{\text{base},i} \quad (13)$$

with hypotheses $H_0 : \mu_d = 0$ and $H_1 : \mu_d \neq 0$. A negative mean difference favours the alternative policy. The KPI values were exported per replication for the analysis and for cross-checking in Minitab.

4 Results and Discussion

This section reports the simulation results for the six policies. Section 4.1 gives the replication-level KPIs and confidence intervals, Section 4.2 the sensitivity analysis, Sections 4.3 and 4.4

the statistical tests, Section 4.5 the effect size, and Section 4.6 the discussion.

4.1 Simulation Results and Policy Comparison

The outputs are stable across the 20 replications. The proposed *Line-aware Slotting* policy reached a mean kitting lead time of 6.91 minutes with a narrow 95% confidence interval of [6.44, 7.38]. Its mean operator waiting time was 4.26 minutes, and reach-truck utilization was only 1.27%, which is consistent with placing frequently used materials in hand-reachable positions. Throughput stayed at about 398.6 orders per day, so the policy improves preparation without reducing output. Tables 5 through 8 report lead time, waiting time, reach-truck utilization, and throughput; Table 9 reports walking distance. Figure 4 compares the policies across four panels.

Table 5: Kitting lead time by policy ($N = 20$ replications).

Policy	Mean (min)	Std. dev.	95% CI
Line-aware Slotting	6.91	0.99	[6.44, 7.38]
Baseline (Actual SAP)	12.37	2.29	[11.30, 13.44]
Baseline (Heuristic)	14.71	2.84	[13.38, 16.04]
Double ABC	14.74	2.89	[13.38, 16.09]
Usage-based ABC	15.24	3.08	[13.80, 16.68]
Travel-distance Optimized	24.60	7.19	[21.23, 27.96]

Table 6: Operator waiting time by policy ($N = 20$ replications).

Policy	Mean (min)	Std. dev.	95% CI
Line-aware Slotting	4.26	0.98	[3.80, 4.72]
Baseline (Actual SAP)	8.45	2.29	[7.38, 9.52]
Baseline (Heuristic)	10.40	2.85	[9.07, 11.73]
Double ABC	10.40	2.90	[9.04, 11.76]
Usage-based ABC	10.81	3.10	[9.37, 12.26]
Travel-distance Optimized	19.03	7.20	[15.66, 22.40]

Table 7: Reach-truck utilization by policy ($N = 20$ replications).

Policy	Mean (%)	Std. dev.	95% CI
Line-aware Slotting	1.27	0.28	[1.14, 1.40]
Baseline (Actual SAP)	21.71	2.46	[20.55, 22.86]
Baseline (Heuristic)	22.82	2.71	[21.55, 24.09]
Double ABC	24.58	2.83	[23.25, 25.90]
Usage-based ABC	24.41	2.74	[23.12, 25.68]
Travel-distance Optimized	43.95	4.72	[41.74, 46.16]

Table 8: Throughput by policy ($N = 20$ replications).

Policy	Mean (orders/day)	Std. dev.	95% CI
Line-aware Slotting	398.58	44.49	[377.76, 419.40]
Baseline (Actual SAP)	398.51	44.45	[377.71, 419.31]
Baseline (Heuristic)	398.38	44.31	[377.64, 419.12]
Double ABC	398.39	44.29	[377.66, 419.12]
Usage-based ABC	398.42	44.42	[377.63, 419.21]
Travel-distance Optimized	396.17	43.16	[375.98, 416.37]

Table 9: Operator walking distance by policy ($N = 20$ replications).

Policy	Mean (m/order)	Std. dev.	95% CI
Baseline (Actual SAP)	89.0	1.30	[88.4, 89.6]
Line-aware Slotting	92.4	1.79	[91.6, 93.3]
Travel-distance Optimized	96.1	1.65	[95.3, 96.9]
Double ABC	104.6	1.70	[103.8, 105.4]
Baseline (Heuristic)	106.1	1.90	[105.2, 107.0]
Usage-based ABC	108.8	1.67	[108.0, 109.6]

The walking-distance results in Table 9 are central to the interpretation. *Line-aware Slotting* walks 92.4 m per order, slightly more than the 89.0 m of the current SAP placement, yet its lead time is about half. The gain therefore does not come from shorter walking; it comes from removing reach-truck calls from the kit path, which is examined in Section 4.6.

4.2 Sensitivity Analysis

A one-at-a-time (OAT) sensitivity analysis was run on nine timing and operational parameters, using the capacity heuristic as the reference policy ($N = 5$ replications per cell, baseline lead time 14.00 minutes). Each parameter was varied by $\pm 20\%$ from its calibrated value while the others were held constant. Table 10 reports the resulting minimum and maximum lead time and the swing, defined as the range divided by the baseline.

Lead time is most sensitive to operator walking speed (38.3% swing) and to the arrival rate (37.4%), followed by the reach-truck lift time per level (18.5%). The remaining parameters move lead time by less than 7%. The conclusions are therefore robust to estimation error in the smaller timing constants. Figure 5 shows the same result as a tornado chart.

A stress test was also run by doubling the arrival rate (twice the system load). Under this heavier load the lead-time advantage of *Line-aware Slotting* over the SAP baseline grows from 5.46 minutes at normal load to 20.7 minutes, which shows that the policy becomes more valuable as the system is pushed harder.

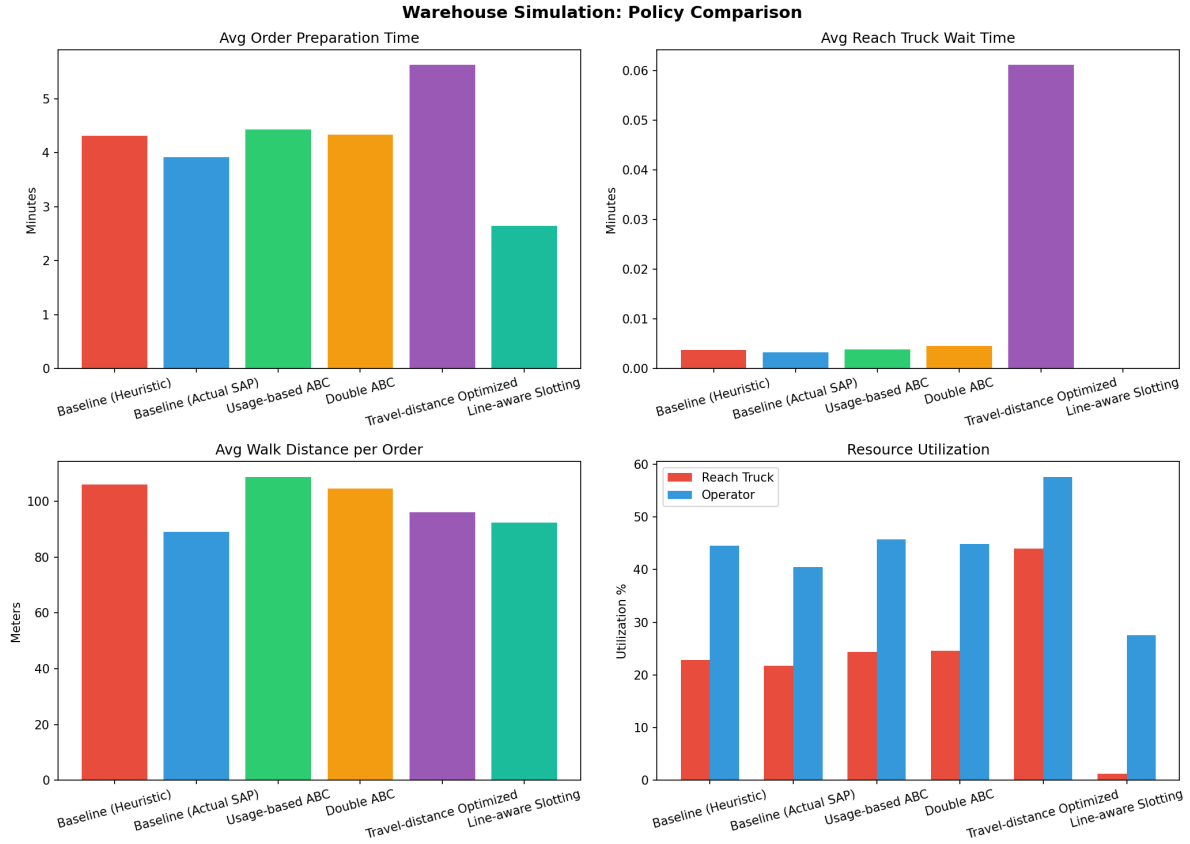


Figure 4: Policy comparison across four panels: average order preparation time, average reach-truck wait time, average walking distance per order, and resource utilization (reach truck and operator). All values are means over $N = 20$ replications with common random numbers.

Table 10: One-at-a-time sensitivity of kitting lead time to $\pm 20\%$ parameter changes (capacity heuristic reference, baseline 14.00 min; $\star$ marks a swing above 15%).

Parameter	Base value	Min lead (min)	Max lead (min)	Swing
Operator walking speed	50 m/min	12.03	17.40	38.3% $\star$
Order inter-arrival time	5.36 min	12.29	17.53	37.4% $\star$
Reach-truck lift time per level	0.25 min	12.74	15.33	18.5% $\star$
Reach-truck travel speed	100 m/min	13.65	14.58	6.6%
Kardex carousel time	0.40 min	13.74	14.27	3.8%
Operator pick time	0.113 min	13.79	14.21	3.0%
Kardex pick time	0.113 min	13.82	14.19	2.6%
Reach-truck pick/place time	0.110 min	13.91	14.11	1.5%
Manual high-level pick penalty	0.102 min	13.99	14.06	0.5%

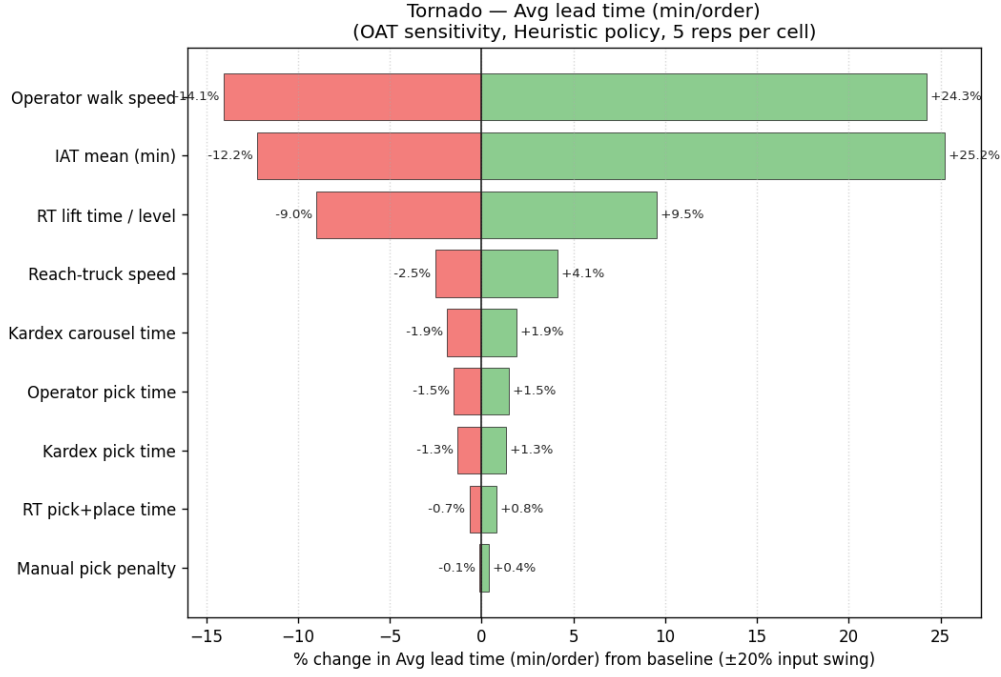


Figure 5: Tornado chart of the one-at-a-time sensitivity of mean kitting lead time to $\pm 20\%$ changes in each parameter.

4.3 Statistical Hypothesis Testing

The one-way ANOVA (Table 11) shows that the choice of storage policy has a significant effect on kitting lead time, waiting time, reach-truck utilization, and walking distance ($p < 0.001$ in each case). Throughput does not differ between policies ($F = 0.01$, $p \approx 1.0$), which confirms that the system is not order-constrained: it completes about the same number of kits per day regardless of policy, so the gains appear as lower lead time and waiting rather than as higher output.

Table 11: One-way ANOVA across the six policies for each KPI ($N = 20$ per policy). H_0 : all policy means are equal.

KPI	F	d.f.	p -value	Decision at $\alpha = 0.05$
Kitting lead time	47.07	(5, 114)	3.55×10^{-26}	Reject H_0
Total waiting time	33.09	(5, 114)	9.86×10^{-21}	Reject H_0
Reach-truck utilization	430.06	(5, 114)	3.19×10^{-72}	Reject H_0
Walking distance	464.15	(5, 114)	5.12×10^{-74}	Reject H_0
Throughput	0.01	(5, 114)	1.000	Fail to reject H_0

4.4 Paired t -Test Results and Policy Contrast

Table 12 reports the common-random-number paired t -tests on kitting lead time against the current SAP placement. *Line-aware Slotting* reduces lead time by 5.46 minutes per order ($t =$

-17.47 , $p = 3.68 \times 10^{-13}$), and the interval does not approach zero. Every other alternative is significantly slower than the current state: the capacity heuristic and the two ABC variants add roughly 2.3 to 2.9 minutes per order, and the travel-distance policy adds 12.23 minutes.

Table 12: Replication-paired t -tests on kitting lead time against the current SAP placement (common random numbers, $N = 20$ pairs). A negative difference favours the alternative policy.

Policy vs. current placement	Mean diff. (min)	t	p -value	Significant?
Capacity heuristic	+2.34	11.27	7.47×10^{-10}	Yes
Usage-based ABC	+2.87	9.97	5.50×10^{-9}	Yes
Double ABC	+2.37	10.05	4.85×10^{-9}	Yes
Travel-distance Optimized	+12.23	10.06	4.82×10^{-9}	Yes
Line-aware Slotting	-5.46	-17.47	3.68×10^{-13}	Yes

The pattern is clear: frequency-based or distance-based slotting alone is not enough for this warehouse. The drivers of improvement are line-based accessibility and a reduced dependence on reach trucks. The *Line-aware Slotting* policy is the only alternative that improves on the current state, because it considers both the production-line relationship and the manual accessibility of materials, which lowers reach-truck intervention and stabilizes the kitting process.

4.5 Practical Effect Size: Cohen’s d

ANOVA and the paired t -tests establish statistical significance; Cohen’s d measures the practical size of the improvement. For the primary contrast, *Line-aware Slotting* versus the SAP baseline, the effect size is $d = 3.10$ (mean difference 5.46 min, pooled standard deviation 1.76 min), as in Equation (14) and Table 13. Since $d \geq 0.8$ is conventionally large, a value of 3.10 indicates a very large practical effect, which confirms that the improvement is operationally meaningful and not just statistically detectable.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (14)$$

Table 13: Cohen’s d for the primary policy contrast (benchmarks: $d \geq 0.2$ small, ≥ 0.5 medium, ≥ 0.8 large).

Contrast	Mean diff. (min)	Pooled SD (min)	Cohen’s d
Line-aware vs. SAP baseline (lead time)	5.46	1.76	3.10

4.6 Discussion

Storage-assignment research has repeatedly found that good slotting can cut order-picking travel by 10 to 30% relative to random or class-based baselines. The present result is consistent in

magnitude, since *Line-aware Slotting* walks 92.4 m per order against 106.1 m for the capacity heuristic, a 13% reduction. It also adds a dimension that the layout literature rarely emphasizes: in a kitting environment with shared reach trucks, the dominant benefit of slotting is the reduction of resource contention rather than of travel distance.

The kitting context is central. A kit order bundles materials from across the warehouse into one set. Limere et al. [17] show that the choice between kitting and line stocking turns on handling cost and distance; this study shows that when reach trucks are shared, the handling cost is mainly queueing time. A policy that ignores this turns a resource that is idle about 78% of the time under SAP placement into one that is saturated at 44% under a pure travel-distance policy. Simulation is well suited to expose this non-linear behaviour, as also noted by Negahban and Smith [18] and Amorim-Lopes et al. [1].

Three features drive the advantage of *Line-aware Slotting*. First, the line-origin distance metric places each material near the line it serves most; because about 77% of kit demand originates at line-specific corridor points, this concentrates picks in small neighbourhoods. Second, the preference for no-reach-truck levels keeps materials at hand height and reserves the fleet for genuine high-level retrieval. Third, by spreading materials across several corridor segments, the policy avoids the local congestion that cripples the travel-distance strategy.

The Usage-based and Double ABC policies both follow the textbook rule to place fast movers nearest the output point. In a single-picker warehouse this is sound, but here there is no single dispatch dock: each line has its own kitting corridor. Collapsing that structure into one distance metric creates the same congestion as the travel-distance policy, only to a smaller degree. The travel-distance policy is the extreme case, optimizing the wrong objective and producing a lead time about double the current state.

For the Manisa plant the implication is direct: re-slot fast movers line by line, with priority to the levels reachable by hand. At about 400 kit orders per day, the model estimates a mean reduction of 5.46 minutes per order, which is a substantial capacity gain over a shift with no new equipment or headcount. One caveat is that the 1.3% reach-truck utilization reported for the proposed policy is a picking-only figure. Putaway and replenishment, which would rise after moving high-turnover items to lower levels, are outside the model; quantifying that trade-off is a priority for future work. The comparison itself is fair, since every policy faces the same demand trajectory under common random numbers, the same resource pool, and the same modelling assumptions.

5 Conclusions and Future Work

5.1 Conclusions

This study developed a discrete-event simulation model of the Schneider Electric Manisa warehouse using historical SAP data, warehouse layout information, Bill of Materials (BOM) re-

lationshps, and operational time studies. The model was used to evaluate alternative storage-assignment policies under realistic operating conditions, and it allowed those policies to be tested without disrupting the ongoing warehouse activity. Six policies were compared in total: the current SAP placement, a capacity heuristic, two frequency-based ABC variants, a travel-distance optimizer, and the proposed *Line-aware Slotting* policy.

The results showed that storage-assignment decisions have a clear and measurable effect on warehouse performance. Among the evaluated alternatives, the proposed *Line-aware Slotting* policy achieved the best overall performance: it reduced mean kitting lead time from 12.37 to 6.91 minutes per order, an improvement of about 44%, while reach-truck utilization fell from 21.7% to 1.3%. One-way ANOVA and common-random-number paired t -tests confirmed that these improvements are statistically significant and are not an artefact of simulation randomness, and the very large effect size ($d = 3.10$) confirms that the gain is operationally meaningful as well.

The findings indicate that the primary source of inefficiency in the current system is not travel distance but the operational dependency created by reach-truck intervention during kit preparation. By positioning frequently used, line-specific materials within hand-reachable storage, the proposed policy reduces waiting time, improves the synchronization between the warehouse and the production lines, and increases overall process efficiency. Notably, it achieves this even though it walks slightly more per order (92.4 m versus 89.0 m for the current placement), which underlines that the lever is queueing rather than distance. Overall, the study demonstrates that a substantial performance improvement is available from a data-driven storage-assignment policy alone, without costly physical layout changes or investment in automation.

5.2 Managerial Implications

The results provide practical guidance for warehouse managers who want to improve operational efficiency with limited investment. The proposed *Line-aware Slotting* policy can be implemented gradually rather than all at once, by first identifying the most frequently used materials of each line and relocating them to manually accessible levels close to the production lines they support. For a facility that processes about 400 kit orders per day, the observed reduction in kitting lead time represents a substantial opportunity to cut waiting waste, improve the utilization of operators and reach trucks, and increase the stability of the kitting process, which in turn reduces the risk of a line-feed disruption. Beyond the specific policy, the simulation framework built in this study is itself a decision-support tool: alternative storage policies can be evaluated and validated virtually before any physical change is made, which lowers both the operational risk and the cost of experimenting on the live warehouse.

5.3 Limitations

While the simulation provides a sound framework for evaluating storage-assignment policies, several modelling choices and scope constraints should be acknowledged, as they define the boundaries within which the results should be interpreted.

Extrapolation of operational times. The operational time parameters were derived from a detailed video time-motion study of 2,319 micro-events conducted specifically on the F400 kitting line, and they were then extrapolated to the other eight product families. If the F400 line is not fully representative of slower or faster assembly lines, the lead-time estimates may deviate accordingly. The sensitivity analysis bounds this risk quantitatively, but line-specific time studies would further refine the model’s accuracy.

Validation data and holdout testing. The model was both fitted and validated against the same historical SAP ZWM92 dataset, because complete historical KPI records were not available for out-of-sample testing. The goodness-of-fit and daily-volume checks therefore serve as rigorous internal-consistency tests rather than true holdout validation against an independent observation period.

SAP decoding and multi-bin splits. Although the model parses the SAP data accurately, only 750 active materials had fully decoded rack-bay-position codes, and 386 materials in the dataset occupy multiple storage bins that the model does not split between forward (active) and reserve assignments. Consequently, the Actual SAP baseline is best understood as a highly accurate SAP-plus-heuristic hybrid rather than a pure bin-for-bin replay of the existing layout.

Physical routing constraints. Physical operator congestion within shared aisles is not modelled, since agents pass through each other freely in the simulation. As a result, the model may slightly underestimate total kitting lead time under the high-load, high-congestion policies, such as the Travel-distance Optimized policy.

Scope limited to order picking. Most importantly, the scope of the simulation is restricted to order picking and kitting. The proposed *Line-aware Slotting* policy concentrates fast-moving materials at lower levels, which will inevitably increase the putaway and replenishment frequency for those rack positions. The reported 1.3% reach-truck utilization for the proposed policy therefore reflects picking workload only, and the total reach-truck workload in a live deployment would be higher once replenishment cycles are included. Potential improvements related to replenishment scheduling, workforce planning, dynamic routing, and equipment allocation were intentionally excluded from this study in order to isolate the specific impact of storage location assignment.

5.4 Future Work

This study can be extended in several directions. First, additional time studies could be carried out across the other production lines to obtain line-specific operational parameters and further improve the accuracy of the model. Second, kit-based co-location strategies, which place com-

ponents that are frequently picked together in adjacent positions, could be integrated with the proposed *Line-aware Slotting* approach to reduce the picking travel within a single order even further. Third, a pilot implementation in a selected warehouse zone would allow a direct before-and-after comparison of real operational performance, providing independent validation of the policy. Finally, the simulation could be combined with optimization techniques such as mathematical programming or metaheuristics to generate and evaluate improved storage-assignment strategies automatically rather than by manual policy design.

5.5 Sustainable Development

This study aligns with three of the United Nations Sustainable Development Goals (SDGs) [23], which shows how operational efficiency can also support sustainable industrial practice. SDG 9 (Industry, Innovation, and Infrastructure) is the most direct fit: moving from intuition-based storage assignment to a data-driven simulation framework is an example of evidence-based industrial innovation, because it improves the process through analysis rather than through new equipment, additional capital expenditure, or facility expansion, which in turn supports resilient and efficient infrastructure. SDG 12 (Responsible Consumption and Production) applies because the proposed *Line-aware Slotting* policy makes fuller use of the storage capacity and material-handling equipment that already exist. By reducing reach-truck utilization, the system consumes less energy per prepared kit, and by improving spatial efficiency it can delay or remove the need for a physical building expansion, which conserves land and construction materials. SDG 8 (Decent Work and Economic Growth) applies on the operator side: shifting high-frequency items into the lower manual band reduces the ergonomic strain associated with mast-elevated retrieval and removes the non-value-added idle time that operators otherwise spend waiting for a reach truck, which contributes to a safer and more productive work environment. These are not transformative global claims; they are concrete, honest improvements that follow directly from the findings of the model.

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A Abbreviations and Nomenclature

B Full KPI Table

Table 15 reports the mean of every KPI over the $N = 20$ replications for all six policies, as a single reference.

Table 14: Abbreviations used in the report.

Abbreviation	Meaning
ABC	Activity-based classification (A/B/C demand classes)
ANOVA	Analysis of variance
BOM	Bill of materials
CRN	Common random numbers
CV	Coefficient of variation
DES	Discrete-event simulation
EPT	Electric pallet truck
FMR	Fast/medium/rare-mover classification
IAT	Inter-arrival time
KPI	Key performance indicator
OAT	One-at-a-time (sensitivity)
RT	Reach truck
SAP	Enterprise resource planning system of the plant
ZWM92	SAP warehouse-management dispatch transaction log

Table 15: Full KPI table, means over $N = 20$ replications.

KPI	Current SAP	Capacity heuristic	Usage ABC	Double ABC	Travel-dist. opt.	Line-aware slotting
Lead time (min)	12.37	14.71	15.24	14.74	24.60	6.91
Prep time (min)	3.92	4.31	4.43	4.34	5.62	2.65
Total wait (min)	8.45	10.40	10.82	10.40	19.03	4.26
P95 lead time (min)	35.9	43.0	42.9	41.9	70.6	18.8
Walk distance (m)	89.0	106.1	108.8	104.6	96.1	92.4
Throughput (orders/day)	398.5	398.4	398.4	398.4	396.2	398.6
RT utilization	0.217	0.228	0.244	0.246	0.440	0.013
Operator utilization	0.405	0.446	0.457	0.448	0.576	0.275
Kardex utilization	0.079	0.079	0.079	0.078	0.078	0.079
Orders completed (5 days)	1943	1942	1942	1942	1931	1943